

Identity-Preserving Video Generation Challenge

Yiheng Zhang
HiDream.ai Inc.
Beijing, China
yihengzhang.chn@hidream.ai

Zhaofan Qiu
HiDream.ai Inc.
Beijing, China
qiuzhaofan@hidream.ai

Qi Cai
HiDream.ai Inc.
Beijing, China
cqcai@hidream.ai

Yehao Li
HiDream.ai Inc.
Beijing, China
liyehao@hidream.ai

Fuchen Long
HiDream.ai Inc.
Beijing, China
longfuchen@hidream.ai

Yingwei Pan
HiDream.ai Inc.
Beijing, China
pandy@hidream.ai

Ting Yao
HiDream.ai Inc.
Beijing, China
tiyao@hidream.ai

Tao Mei
HiDream.ai Inc.
Beijing, China
tmei@hidream.ai



Figure 1: Eight samples of the VIP-200K. For each unique identity, two face images and two video clips are shown.

Abstract

Recent advancements in multimodal AIGC have enabled impressive text-to-video synthesis, but a critical challenge remains: maintaining consistent identity of key subjects across generated frames. To address this limitation, we introduce the Identity-Preserving Video Generation (IPVG) grand challenge. This challenge aims to propel the field toward more controllable generative models by focusing community efforts on preserving identity during the video generation process. To support these efforts, we publicly release the Identity-Preserving Video Benchmark (VIP-200K), a

novel dataset comprising approximately 500,000 video-prompt pairs with 200,000 unique identities, each coupled with a reference identity image. Through this grand challenge and dataset, we provide a fertile ground for developing solutions that lead to more user-steerable video synthesis systems. The challenge homepage is <https://hidream-ai.github.io/ipvg-challenge.github.io/>.

CCS Concepts

• Information systems → Multimedia content creation.

Keywords

Dataset, Identity-Preserving, Video Generation, AIGC

ACM Reference Format:

Yiheng Zhang, Zhaofan Qiu, Qi Cai, Yehao Li, Fuchen Long, Yingwei Pan, Ting Yao, and Tao Mei. 2025. Identity-Preserving Video Generation Challenge. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3746027.3761987>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3761987>

Table 1: Comparison of VIP200K with existing open-source datasets for human-centric identity-preserving video generation. † indicates that only the publicly available portion of the dataset is considered.

Dataset	#Samples	#IDs	Resolution	Duration (hours)	Source	OpenSource
FaceForensics [20]	1,004	1,004	256×256	4	Wild	Yes
VoxCeleb [17]	21,245	1,251	224×224	352	Wild	Yes
VoxCeleb2 [5]	150,480	6,112	224×224	2,442	Wild	Yes
MEAD [24]	281,400	60	1080p	39	Lab	Yes
CelebV-HQ [31]	35,666	15,653	512×512	68	Wild	Yes
CelebV-Text [29]	70,000	-	512×512	279	Wild	Yes
ConsisID [†] [30]	31,947	31,947	720p	41	Wild	Yes
VIP-200K	500,000	200,000	720p ~ 4k	1,700	Wild	Yes

1 Introduction

Recent advancements in Multimodal Artificial Intelligence-Generated Content (AIGC) [1, 4, 19, 26, 32] have catalyzed a paradigm shift in digital content creation. Cutting-edge models such as Stable Diffusion [19], Flux [1], and HiDream-I1 [2] have demonstrated unprecedented capabilities in synthesizing high-fidelity images from textual prompts, while systems like Sora, Runway Gen-3, and HiDream.ai have extended these breakthroughs to the video domain. The emergence of such technologies has not only astonished the research community but also underscored the transformative potential of AI-driven video generation across industries, including entertainment, education, virtual reality, and personalized media.

Video generation [7, 18], as a fundamental task in multimedia field, demands the simultaneous modeling of spatial details, temporal dynamics, and semantic consistency. State-of-the-art text-to-video systems excel at producing visually compelling content directly from textual prompts. However, a critical limitation persists: the inherent stochasticity of diffusion processes often leads to inconsistent appearances of key subjects (e.g., characters, objects) across frames/scenes. For instance, facial features of a character may drift unnaturally over time, or object attributes might degrade during prolonged sequences. This instability stems from inherent challenges in disentangling identity-related features from contextual variables (e.g., pose, lighting, occlusion) within high-dimensional latent spaces—a problem exacerbated by the stochastic nature of generative processes. In this grand challenge, we aim to alleviate this limitation by introducing Identity-Preserving Video Generation (IPVG) task, which maintains the consistency of given reference identity along text-to-video generation process. Practically, IPVG is pivotal for applications such as personalized avatar animation, historical figure reenactment, and brand-specific content creation, where even minor identity deviations can compromise usability or authenticity.

This year’s IPVG grand challenge includes one tracks: Facial Identity-Preserving Video Generation. In general, the goal of this grand challenge is two-fold: (a) coalescing community effort around a new challenging identity-preserving video generation dataset, and (b) offering a fertile ground for designing controllable generative models to facilitate precise identity binding in video generation, aiming to propel the field toward more accountable and user-steerable video synthesis systems.

2 Related Work

Identity-Preserving Image/Video Generation. This task aims to create images and/or videos that faithfully incorporate reference content (i.e. ID), providing enhanced flexibility in the production of AIGC. Early approaches, such as DreamBooth [21] and Textual Inversion [8], rely on optimization-based methods: DreamBooth fine-tunes the entire text-to-image model for each subject, while Textual Inversion learns subject-specific text embeddings to guide novel image generation. However, these methods typically require per-subject finetuning, resulting in slow adaptation and susceptibility to overfitting. To address these issues, recent research has explored encoder-based techniques that reduce or eliminate test-time finetuning. For instance, IP-Adapter [28] introduces a lightweight, decoupled cross-attention mechanism for image conditioning, and Instantbooth [22] leverages an image encoder to map reference images to textual tokens, supplemented by adapter layers for identity preservation. Such techniques are also compatible with video generation. Models like DreamVideo [25] train image/motion adapters to capture the appearance and motion of subjects. In real-world applications, identity-preserving generation is particularly valuable in scenarios where humans are the central subjects—such as personalized avatars, digital doubles, and video editing for social media or film. For example, methods like PhotoMaker [16] and PuLID [9] are specifically designed for portrait image generation, achieving impressive results in maintaining facial characteristics. Similarly, in the video domain, works such as ID-Animator [11] and ConsisID [30] have demonstrated strong performance in generating videos with high facial fidelity and temporal consistency.

Datasets for Identity-Preserving Video Generation. The advancement of diffusion models, especially for human-centric identity-preserving video generation, is heavily dependent on the availability of large-scale, high-quality datasets. Comprehensive datasets not only enable the training of more robust generative models but also provide reliable benchmarks for fair and consistent evaluation. As shown in Table 1, several open-source video datasets centered on human subjects have been released to the academic community. It reveals that most existing datasets fall short in either scale or quality, limiting their effectiveness for driving progress in diffusion-based generation. In particular, many datasets lack diversity, high-resolution video, or sufficient annotation, which poses



Figure 2: Overview of the VIP-200K Dataset Construction Pipeline. The process consists of four main steps: **DataCuration**, **Temporal Filtering**, **Human Identity Detection**, and **PostProcessing**. In **DataCuration**, query words related to humans are first collected to retrieve full-length videos, which are then filtered based on resolution, duration, content type, and the presence of humans. The filtered videos are subsequently segmented by shot transitions, and clips are further refined based on content and quality. In **Temporal Filtering**, a YOLO detection model is used to perform frame-by-frame human detection, followed by segmentation to ensure that clips retain human presence for the majority of their duration. The **Human Identity Detection** step employs face analysis tools to extract and cluster facial features, assigning human IDs and associating video clips with each ID, ultimately forming the dataset.

significant challenges for developing and evaluating state-of-the-art identity-preserving video generation methods. Consequently, there is a pressing need for new, high-quality benchmarks tailored to the unique demands of this rapidly evolving research area.

To address the lack of high-quality open-source datasets, we present VIP-200K, a large-scale benchmark focused on human-centric identity-preserving video generation. VIP-200K contains 500,000 high-resolution video clips, representing 200,000 unique human identities, each paired with descriptive captions. This dataset provides the scale and diversity needed for advancing identity-preserving video generation. VIP-200K has served as the benchmark in our Identity-Preserving Video Generation Challenge and its effectiveness has been validated.

3 VIP-200K

In this section, we provide a comprehensive overview of the VIP-200K dataset, which serves as the foundation for the Identity-Preserving Video Generation Challenge. We begin by detailing the data pipeline in section 3.1, including the methods and workflow used to construct the dataset. Next, in section 3.2 we present key data statistics that highlight the diversity and distribution of the dataset across various dimensions. Finally, we describe the evaluation metrics and benchmarks used for assessing model performance,

including the design of the validation set and the selection of relevant metrics in section 3.3.

3.1 Data Pipeline

To curate the VIP-200K dataset, we followed a multi-step pipeline to ensure both the quality and diversity of the video clips, as demonstrated in Figure 2.

Data Curation. Initially, we compiled a set of 1,000 human-centric keywords by selecting terms from several categories to ensure broad coverage of human-related subjects. These categories included personal pronouns (e.g., “he,” “she,” “they,” and gender-neutral terms like “them” and “ze”), general human-related nouns such as “man,” “woman,” “boy,” “girl,” “baby,” and “child” to represent different age groups and genders, and common professions like “doctor,” “teacher,” “engineer,” “chef,” and “scientist” to capture a variety of occupational identities. Additionally, we included terms related to social roles and relationships, such as “parent,” “student,” and “elderly,” to reflect diverse human dynamics and contexts. Finally, we incorporated activity and context-based terms like “dancer,” “athlete,” “cooker,” and “speaker” to represent various actions and scenarios. By gathering query words from these diverse categories, we ensured a comprehensive set of human-centric keywords that cover both individual identities and the various contexts in which humans exist.

Using these keywords, we performed web scraping on various video platforms, including YouTube, to collect videos that met a set of strict criteria. Initially, we conducted a quality filter using meta-information such as the number of views, likes, and the number of subscribers of the video creator. This helped to discard videos with low engagement and questionable content quality. Next, we applied further filters based on video resolution and duration, retaining only videos with a resolution of at least 720p and a duration of no less than 5 seconds. In cases where multiple versions of the same video were available, we retained only the version with the highest resolution and frame rate. After applying these filters, we obtained a total of 460K full videos.

To coarsely filter out irrelevant content, we employed an off-the-shelf vision-language model (MiniCPM-V [27]) within a question-answering framework to automatically assess whether each video depicted a real-world scene and included at least one human subject. Specifically, for each video, we sampled four evenly spaced segments, each lasting two seconds, in addition to the complete video, resulting in five representative clips per video. To eliminate animated or cartoon content, we posed the question “Is this video animated or cartoon? Answer YES or NO.” to the VLM for each of the five clips. If more than one response was “YES,” the video was deemed non-realistic and subsequently discarded. To further ensure human presence, we queried the VLM with “Are there any humans in the video? Answer YES or NO.” across all five clips, requiring at least one affirmative response. Videos failing to meet this criterion were excluded from further consideration. This two-stage VLM validation process effectively removed irrelevant and non-human-centric videos, ensuring that the resulting dataset was composed of real-world footage featuring visible human subjects.

The remaining videos were segmented into individual shots using PySceneDetect [3], a tool specifically designed to automatically detect shot changes within video streams. By identifying scene transitions, PySceneDetect splits each video into separate, temporally coherent clips, resulting in an initial pool of more than 2 million video segments. Subsequently, we applied a suite of automated video quality assessment tools to evaluate each segment across multiple dimensions, including visual clarity, motion dynamics, and aesthetic quality. Segments receiving low scores in any of these categories were discarded. After this stringent quality control procedure, we retained 1.5 million high-quality, human-centric video clips, which served as the candidate pool for the subsequent stages of dataset refinement and filtering.

Temporal Filtering. Although the initial data curation pipeline effectively filters for high-quality, human-centric video clips, additional refinement is necessary to further enhance dataset suitability for identity-preserving video generation tasks. Empirically, videos used for training such models must not only feature human subjects but also minimize the proportion of frames unrelated to human activity. However, web-sourced video data, unlike professionally edited cinematic content, often includes substantial segments with irrelevant or distracting content. Despite previous filtering steps, it remains challenging to guarantee a consistently high ratio of person-centric frames within each segment. Therefore, we introduce a temporal filtering process to further improve data relevance and quality.

For each candidate video clip, we apply the following temporal filtering procedure: (1) The YOLOv11x detection model is used to perform frame-by-frame human detection, with the position and size of each detected person recorded. (2) For every frame, we calculate the area of each detected bounding box and remove detections whose area ratio relative to the frame is below a predefined threshold, thereby eliminating small or peripheral detections. (3) Based on the number of valid human regions in each frame, we label the frame as containing humans if it includes one or more valid detections; otherwise, it is marked as not containing humans. (4) Using the resulting sequence of binary human-presence labels, we scan forward from the first frame to identify the start and end points of valid segments. A segment is defined as valid if its first and last frames both contain humans, no humans are present before the start or after the end of the segment, there are at most five consecutive frames without human presence within the segment, and the total length is at least 120 frames.

After temporal filtering, each valid video segment contains at least one human subject, and the presence of the subject occupies the majority of the segment’s duration. Each valid segment is treated as an independent clip for subsequent processing steps. As a result of this procedure, we obtained approximately 1 million valid video clips.

Human Identity Detection. A Human Identity (ID) is defined as a unique and consistent individual who may appear across multiple video clips within the dataset. Accurate identification and annotation of these identities are important for supporting identity-preserving video generation tasks. Cross-clip identity matching is necessary because, in many cases, conditioning images and target videos sampled from the same clip (intra-clip pairs) exhibit high visual similarity, which can lead to trivial solutions during model training and limit the diversity of training examples. In contrast, matching identities across different clips (inter-clip pairs) introduces greater variation in appearance, which encourages the model to learn more robust and generalizable representations for identity preservation. The aim of this stage is to extract and assign unique identity labels to individuals within the pool of video clips, and to systematically record the appearance of each identity across the relevant clips for use in subsequent training and evaluation.

To extract human identity information efficiently and reliably, we first organize the video clips into groups, where each group consists of clips derived from the same full-length video. This grouping is based on the reasonable assumption that the set of individuals appearing within a single source video is limited and relatively stable. Restricting inter-clip identity retrieval to within these groups reduces computational complexity while maintaining accuracy. Within each group, we utilize the InsightFace face analysis toolkit to detect faces at 16-frame intervals across all clips. For every detected face, we extract facial attributes and embedding features, which serve as the basis for subsequent identity clustering. Low-quality detections, such as faces that are too small or detected with low confidence, are removed to ensure that only reliable samples are considered. Finally, we perform clustering on the extracted face embeddings using the DBSCAN algorithm, requiring that each cluster contains at least five faces to ensure cluster robustness. Each resulting cluster is treated as a distinct human identity (ID), and we record both the representative face samples and the associated

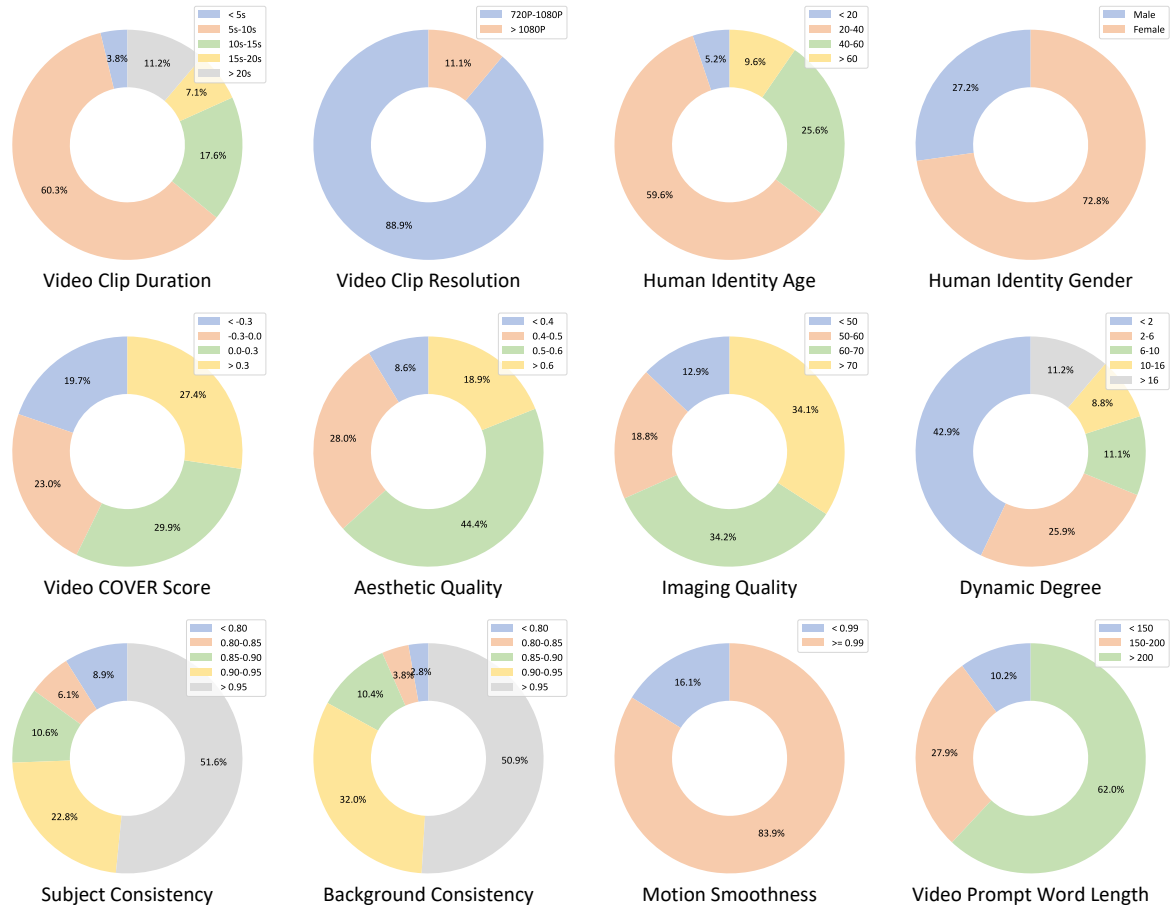


Figure 3: Statistical overview of the VIP-200K dataset. The figure summarizes the distribution of video clip durations and resolutions, VBench quality scores, COVER scores, attributes of human identities (age and gender), and the length of captions.

video clips for each ID. This process enables accurate cross-clip identity annotation and lays a solid foundation for downstream tasks in identity-preserving video generation.

After the Human Identity Detection process, we obtained a total of 200,000 valid identities (IDs), which are distributed across 500,000 video clips. Based on these annotated identities and their corresponding video segments, we constructed the VIP-200K dataset.

Post Processing. In the final stage of data preparation, each video clip in the dataset undergoes detailed post processing to enhance its usability for research and application. To provide rich, human-centric annotations, we employ a vision-language model (MiniCPM-V) to generate descriptive captions for every video clip. The model is instructed to focus its descriptions specifically on the person or people appearing in the clip, providing detailed accounts of their physical appearance, clothing, posture, and actions, as well as the surrounding environment. This approach ensures that the generated captions maintain a strong emphasis on human-related content and context, which is essential for downstream identity-preserving video generation tasks.

In addition to comprehensive annotation, we take further steps to facilitate the dissemination and use of the VIP-200K dataset within the open-source community. For each video clip, we provide not

only the original video URL and the corresponding annotation information, but also a processed version of the video. The processed videos are uniformly scaled so that the shorter side is 480 pixels, and are encoded using the H.265 codec to achieve a compact file size while preserving visual quality. This post-processing ensures that the dataset remains accessible and practical for large-scale research while maintaining the richness and quality of the content.

3.2 Data Statistics

This section provides a systematic analysis of the VIP-200K dataset, summarizing its statistical properties and examining the distribution of key attributes from multiple perspectives. The analyses include several aspects relevant to the dataset’s potential use in identity-preserving video generation and related tasks. The statistical results are illustrated in Figure 3.

We first report the basic properties of the video clips, including distributions of duration and resolution, to illustrate the dataset’s diversity in scale and temporal characteristics. The overall quality of the video clips is assessed using six evaluation metrics adopted from VBench [15]: subject consistency, background consistency, motion smoothness, dynamic degree, aesthetic quality, and imaging

quality. In addition, we present results for the COVER [10] score, a completely blind video quality evaluation metric, to further quantify video fidelity. The section also includes an analysis of identity attributes, such as age and gender, providing an overview of the demographic diversity among annotated human identities. Lastly, we analyze the length distribution of the generated captions.

3.3 Challenge Evaluation

This section describes the construction of evaluation data, and metrics used in the VIP-200K Challenge to assess submissions from participating teams. Both objective and subjective measures are adopted to ensure a comprehensive evaluation.

Evaluation Set. To construct the validation set, we first collected video clips that were excluded during the **Human Identity Detection** stage. We removed clips originating from the same full video as any training set clip, ensuring a clear separation between training and validation data. For the remaining clips, we performed face and body detection to identify frames containing human subjects. Facial embeddings were then compared to all training set identities, and only those faces not present in the training set were retained. From these, 200 samples were randomly selected, each from a different full video, to serve as validation identities. For text conditioning, we curated a pool of 100 prompts using a “profession + action” template, avoiding references to gender, age, or ethnicity. Each validation identity was paired with four sampled prompts from this pool, as well as the descriptive caption of its source video clip, resulting in five prompts per identity. In total, the validation set comprises 1,000 ID, prompt pairs, enabling robust and fair evaluation of identity-preserving video generation models.

Objective Metrics. For identity preservation, we use FaceSim-Arc and FaceSim-Cur, which measure feature differences between face regions in generated videos and real face images using the ArcFace [6] and CurricularFace [14] embedding spaces. Visual quality is assessed with the Fréchet Inception Distance (FID) [13], computed on face regions in the InceptionV3 [23] space, as well as an Aesthetic Score reflecting visual appeal. Text relevance is measured by CLIPScore [12], which quantifies the alignment between generated videos and input prompts.

Subjective Evaluation. For top submissions, a user study was conducted. For each test case, pairwise comparisons between team results were randomly assigned to annotators, who selected the preferred sample based on identity preservation, video quality, and prompt relevance. Wins, ties, and losses were scored as 1, 0.5, and 0, respectively, and team rankings were determined by overall scores.

Acknowledgments. This work was supported in part by the Beijing Municipal Science and Technology Project No. Z241100001324002 and Beijing Nova Program No. 20240484681.

References

- [1] Stephen Batifol, Andreas Blattmann, Frederic Boesel, et al. 2025. FLUX. 1 Kontext: Flow Matching for In-Context Image Generation and Editing in Latent Space. *arXiv e-prints* (2025), arXiv–2506.
- [2] Qi Cai, Jingwen Chen, Yang Chen, Yehao Li, et al. 2025. HiDream-1I: A High-Efficient Image Generative Foundation Model with Sparse Diffusion Transformer. *arXiv preprint arXiv:2505.22705* (2025).
- [3] Brandon Castellano. 2025. PySceneDetect: Video Cut Detection and Analysis Tool. <https://github.com/Breakthrough/PySceneDetect>
- [4] Jingwen Chen, Yingwei Pan, Ting Yao, and Tao Mei. 2023. Controlstyle: Text-driven stylized image generation using diffusion priors. In *ACM Multimedia*.
- [5] J. S. Chung, A. Nagrani, and A. Zisserman. 2018. VoxCeleb2: Deep Speaker Recognition. In *INTERSPEECH*.
- [6] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. Arcface: Additive angular margin loss for deep face recognition. In *CVPR*.
- [7] Ting Yao Fuchen Long, Zhaofan Qiu and Tao Mei. 2024. VideoStudio: Generating Consistent-Content and Multi-Scene Videos. In *ECCV*.
- [8] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. 2023. An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion. In *ICLR*.
- [9] Zinan Guo, Yanze Wu, Zhuowei Chen, Lang Chen, Peng Zhang, and Qian He. 2024. PuLID: Pure and Lightning ID Customization via Contrastive Alignment. In *NeurIPS*.
- [10] Chenlong He, Qi Zheng, Ruoxi Zhu, Xiaoyang Zeng, Yibo Fan, and Zhengzhong Tu. 2024. COVER: A Comprehensive Video Quality Evaluator. In *CVPR Workshops*.
- [11] Xuanhua He, Quande Liu, Shengju Qian, Xin Wang, Tao Hu, Ke Cao, Keyu Yan, and Jie Zhang. 2024. Id-animator: Zero-shot identity-preserving human video generation. *arXiv preprint arXiv:2404.15275* (2024).
- [12] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. 2021. Clipscore: A reference-free evaluation metric for image captioning. *arXiv preprint arXiv:2104.08718* (2021).
- [13] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *NeurIPS*.
- [14] Yuge Huang, Yuhang Wang, Ying Tai, Xiaoming Liu, Pengcheng Shen, Shaoxin Li, and Feiyue Huang Jilin Li. 2020. CurricularFace: Adaptive Curriculum Learning Loss for Deep Face Recognition. In *CVPR*.
- [15] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. 2024. VBench: Comprehensive Benchmark Suite for Video Generative Models. In *CVPR*.
- [16] Zhen Li, Mingdeng Cao, Xintao Wang, Zhongang Qi, Ming-Ming Cheng, and Ying Shan. 2024. PhotoMaker: Customizing Realistic Human Photos via Stacked ID Embedding. In *CVPR*.
- [17] A. Nagrani, J. S. Chung, and A. Zisserman. 2017. VoxCeleb: a large-scale speaker identification dataset. In *INTERSPEECH*.
- [18] Yingwei Pan, Zhaofan Qiu, Ting Yao, Houqiang Li, and Tao Mei. 2017. To create what you tell: Generating videos from captions. In *ACM Multimedia*.
- [19] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *CVPR*.
- [20] Andreas Rössler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. 2019. FaceForensics++: Learning to Detect Manipulated Facial Images. In *ICCV*.
- [21] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. 2022. DreamBooth: Fine Tuning Text-to-image Diffusion Models for Subject-Driven Generation. *arXiv preprint arXiv:2208.12242* (2022).
- [22] Jing Shi, Wei Xiong, Zhe Lin, and Hyun Joon Jung. 2024. Instantbooth: Personalized text-to-image generation without test-time finetuning. In *CVPR*.
- [23] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. 2016. Rethinking the inception architecture for computer vision. In *CVPR*.
- [24] Kaisiyan Wang, Qianyi Wu, Linsen Song, Zhuoqian Yang, Wayne Wu, Chen Qian, Ran He, Yu Qiao, and Chen Change Loy. 2020. MEAD: A Large-scale Audio-visual Dataset for Emotional Talking-face Generation. In *ECCV*.
- [25] Yujie Wei, Shiwei Zhang, Zhiwu Qing, Hangjie Yuan, Zhiheng Liu, Yu Liu, Yingya Zhang, Jingren Zhou, and Hongming Shan. 2024. DreamVideo: Composing Your Dream Videos with Customized Subject and Motion. In *CVPR*.
- [26] Ting Yao, Yehao Li, Yingwei Pan, Zhaofan Qiu, and Tao Mei. 2025. Denoising token prediction in masked autoregressive models. In *ICCV*.
- [27] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, et al. 2024. MiniCPM-V: A GPT-4V Level MLLM on Your Phone. *arXiv preprint arXiv:2408.01800* (2024).
- [28] Hu Ye, Jun Zhang, Sibio Liu, Xiao Han, and Wei Yang. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. *arXiv preprint arXiv:2308.06721* (2023).
- [29] Jianhui Yu, Hao Zhu, Liming Jiang, Chen Change Loy, Weidong Cai, and Wayne Wu. 2023. CelebV-Text: A Large-Scale Facial Text-Video Dataset. In *CVPR*.
- [30] Shenghai Yuan, Jinfa Huang, Xianyi He, Yunyang Ge, Yujun Shi, Liuhan Chen, Jiebo Luo, and Li Yuan. 2025. Identity-preserving text-to-video generation by frequency decomposition. In *CVPR*.
- [31] Hao Zhu, Wayne Wu, Wentao Zhu, Liming Jiang, Siwei Tang, Li Zhang, Ziwei Liu, and Chen Change Loy. 2022. CelebV-HQ: A Large-Scale Video Facial Attributes Dataset. In *ECCV*.
- [32] Rui Zhu, Yingwei Pan, Yehao Li, Ting Yao, Zhenglong Sun, Tao Mei, and Chang Wen Chen. 2024. Sd-dit: Unleashing the power of self-supervised discrimination in diffusion transformer. In *CVPR*.